The Preferendus Landscape

An integrated ecosystem for real-world Open Design and Decision Systems (ODESYS)

Executive Summary

ODESYS-in-FIVES is an open design and decision systems methodology for confronting conflicts of interest within real-world complexity. Its primary application domains are systems design and engineering management, supported by preference-performance optimisation and computational methods.

It provides a multidimensional framework for multi-actor preference function modelling, multi-level performance modelling, and best-fit solver algorithms, integrating design and decision optimisation and analysis across stakeholders, system scales, and management levels.

It supports those involved in complex design and decision-making across socio-technical systems, where multiple actors, interests, system possibilities, and management levels interact. By bridging what is preferred with what is possible, ODESYS-in-FIVES enables actionable and scalable best-fit solutions, particularly where complexity exceeds what human reasoning alone can effectively manage.

Rather than seeking sub-optimal compromises, ODESYS-in-FIVES aims for best-fit syntheses that maximise aggregated preference value for the whole. To achieve this, ODESYS-in-FIVES identifies the best fit between individual actors' freedom and systemic degrees of freedom, supported by Preferendus, a neutral computational design-decision-support engine.

Structured according to the FIVES paradigm, ODESYS-in-FIVES integrates performance, comprising (1) capability and (2) feasibility; preference, comprising (3) desirability and (4) acceptability; and (5) solvability within a common preference-performance theoretical framework. This framework spans the design-decision problem-solving journey from intent to decisive impact.

ODESYS-in-FIVES introduces two novel and complementary preference-consistent methodologies: **IMAP** for multi-objective design optimisation (MODO), supported by the **IGS Solver Family**, and **TOMAP** for multi-criteria decision analysis (MCDA). Performance information can be represented across four levels of modelling complexity, ranging from directly assigned performance values at Level 0 to advanced simulation models at Level 3. At the core of both methodologies, the A-FINE Aggregator derives decision-valid aggregated preferences within a unified preference-performance domain.

These methodologies, methods, and algorithms are operationalised within the **Preferendus Software Suite**, with **Preferendus-Agent** as its central orchestration architecture. Equipped with ODESYS-in-FIVES skills, its LLM-based **Dialogue Agent** works with problem owners and stakeholders to translate POIs from natural language into ODESYS-in-FIVES blueprints. Its agentic software-engineering **Delivery Agent** then develops these blueprints into working prototypes, operational applications, and scalable design-decision-support solutions. Together, these agents enable native support for Level 0 and Level 1 performance representations while connecting to application-specific Level 2 and Level 3 performance models.

Through an open-ended design approach, ODESYS-in-FIVES purposefully transforms current realities into desired outcomes within socially and physically feasible boundaries. It maps and aggregates individual actor preferences over their weighted performances of interest to identify, from the feasible possibilities, the solution with the highest

aggregated preference value: a best fit for and within the whole. In doing so, it creates an open space for constructive dialogue and brings a common ideal within reach.

Conventional systems engineering, operations research, and multi-criteria design-decision methods provide valuable support, but necessarily reduce real-world complexity to what their models and assumptions can represent. Without preference-consistent integration of stakeholder interests and system performance, valuable design freedom may remain unrecognised and best-fit solutions undiscovered. ODESYS-in-FIVES takes the next step through a co-creative response that truly opens the design space and turns decision-making right side up. It integrates systems thinking, design thinking, social thinking, and slow thinking into a synergistic whole: a deliberate art of design and decision-making aimed at revealing what might otherwise remain unexplored and determining what fits best for the whole.

Together, ODESYS-in-FIVES and the Preferendus Software Suite provide a unified ecosystem for real-world design-decision optimisation, supported by human judgement, computational intelligence, and AI-driven orchestration to deliver actionable and scalable best-fit-for-common-purpose solutions within reach.

Architecture Overview

Layer	Method / Technology	Preferendus Product	Role
Design & Decision Systems	ODESYS-in-FIVES	Preferendus-Agent	Orchestrates ODESYS methods, including IMAP and TOMAP, and their capabilities within a unified AI-driven design and decision environment. It natively supports performance Levels 0 and 1 and can connect to enterprise-specific Level 2 and Level 3 models.
Multi-Objective Design Optimization (MODO)	IMAP: Integrative Maximization of Aggregated Preferences	Preferendus-IMAP-IGS	Performs preference-performance-driven design optimization through IMAP, the IGS solver family, and A-FINE.
Best-Fit Candidate Search	IGS Solver Family	Preferendus-IMAP-IGS	Searches complex design spaces for best-fit-for-common-purpose solutions.
Multi-Criteria Decision Analysis (MCDA)	TOMAP: Trade-Off Matrix by Aggregated Preferences	Preferendus-TOMAP	Produces preference-based trade-off matrices, rankings, analyses, and design-decision support.
Preference Aggregation	A-FINE Aggregator, the core aggregation algorithm	Preferendus-ALL	Derives decision-valid aggregated preferences in the Linear Preference Space (LPS).

ODESYS Methodology Stack

- **A-FINE Aggregator:** Core preference-aggregation algorithm that transforms diverse preference functions into a uniquely determined, preference-consistent, and decision-valid aggregate outcome.
- **IGS Solver Family:** Family of intergenerational solvers for best-fit candidate search in complex MODO design spaces.
- **IMAP:** Integrative Maximization of Aggregated Preferences, the ODESYS methodology for preference-performance-driven MODO.
- **TOMAP:** Trade-Off Matrix by Aggregated Preferences, the ODESYS methodology for preference-based MCDA.
- **ODESYS-in-FIVES:** The skills-based framework that structures performance, preference, and solvability and enables AI-assisted design and decision support.

Preferendus Software Suite

A-FINE Aggregator

An operational preference-aggregation procedure for applied MCDM. It aggregates commensurable preference information through a weighted centroid in the Linear Preference Space (LPS), which provides a stable zero-referenced interval scale and preserves uniqueness up to affine transformation, thereby enabling preference-consistent design–decision analysis and optimisation. Within the contemporary Preferendus software environment, the A-FINE Aggregator is implemented in the IMAP–IGS Solver Family for multi-objective design–decision optimisation (MODO) and in TOMAP for preference-based trade-off design–decision analysis (MCDA).

Preferendus-IMAP-IGS

A multidimensional, multi-objective design-decision optimisation (MODO) platform implementing IMAP (Integrative Maximization of Aggregated Preferences) through the IGS Solver Family and the A-FINE Aggregator. It integrates multi-actor preference functions, multi-level system performance models, and best-fit solver algorithms to explore complex design spaces and discover preference-consistent, best-fit-for-common-purpose solutions using A-FINE's unique decision-valid preference aggregation.

Preferendus-TOMAP

A multidimensional, multi-criteria design-decision analysis (MCDA) tool implementing TOMAP (Trade-Off Matrix by Aggregated Preferences) through the A-FINE Aggregator. It integrates multi-actor preference functions and multi-level system performance information to evaluate predefined alternatives using A-FINE's unique decision-valid preference aggregation. It produces preference-consistent trade-off matrices and transparent design-decision support.

Preferendus-Agent

An AI-driven design and decision agent equipped with ODESYS-in-FIVES skills, orchestrating IMAP-IGS and TOMAP capabilities within an integrated design-decision problem-solving environment. Preferendus-Agent comprises two complementary agents:

- **Dialogue Agent**: an LLM-based conversational interface through which problem owners and stakeholders jointly explore, frame, and translate a real-world Problem of Interest (POI) from natural language into an ODESYS-in-FIVES blueprint. The blueprint includes an initial IMAP-generated best-fit solution and a TOMAP-based relative ranking and trade-off analysis against other relevant alternatives.
- **Delivery Agent**: a human-in-the-loop, agentic software-engineering environment built around GitHub, Visual Studio Code, and GitHub Copilot. It supports experienced ODESYS modellers in extending the blueprint with Level 1–3 performance models, selecting and configuring the IGS solver best suited to the POI, and developing the resulting ODESYS-in-FIVES specification into a working prototype, an operational application, and a scalable design-decision-support solution that delivers actionable results.

Performance Modelling Levels

Performance information can be represented within Preferendus-Agent at four levels. These levels distinguish how performance information is represented, the complexity of the underlying model, and how the information is made available to the design and decision process. Level 0 uses discrete raw performance values for a set of predefined, feasible, and acceptable alternatives. Levels 1 to 3 provide performance functions or models that generate performance values for variable design inputs, using generic analytical equations, enterprise-specific adaptations or surrogate models, and advanced behavioural or simulation models, respectively.

Preferendus-Agent natively supports Level 0 and Level 1 performance representations, making Preferendus-Agent directly applicable to a wide range of design and decision problems. Enterprise-specific Level 2 and Level 3

capabilities can be made available to Preferendus-Agent through dedicated AI skills, tools, model interfaces, and software connectors. Preferendus-Agent orchestrates these levels and transforms their performance outputs into preference-meaningful and decision-valid design optimisation and decision support.

Level 0: Discrete Raw Performance Values

Discrete performance values are available for a finite set of predefined, feasible, and acceptable alternatives. These values may originate from measurements, calculations, expert judgement, historical data, or external models. Within Preferendus-Agent, they are treated as given values and evaluated through preference functions.

Examples: cost, time, weight, capacity, emissions, reliability, catalogue specifications, or previously calculated results.

Level 1: Generic Analytical Equations

Transparent and generally applicable heuristics and mathematical, physical, engineering, economic, or social-science equations that calculate performance as a function of design-decision parameters. These models enable Preferendus-Agent to evaluate performance beyond a predefined set of alternatives.

Examples: calculation rules, engineering formulas, cost functions, or physical relationships.

Level 2: Enterprise-Specific and Surrogate Models

Calibrated, empirical, measurement-based, statistical, reduced-order, or machine-learning models represent enterprise-specific performance. They may adapt generic Level 1 equations or approximate detailed Level 3 simulations.

Examples: enterprise-specific performance curves, regression models, response surfaces, Gaussian processes, reduced-order models, or trained machine-learning models.

Level 3: Advanced Behavioral and Simulation Models

High-fidelity models and specialist software calculate performance by representing detailed physical, dynamic, operational, or organisational system behaviour. They may be connected directly to Preferendus-Agent or used to generate data for Level 2 models.

Examples: discrete-event simulation (DES) models, graph-based models, digital twin models, and spatially discretised simulation models, such as Finite Element Method (FEM) models and Computational Fluid Dynamics (CFD) models, as well as other specialised enterprise performance software.

Level	Representation in Preferendus-Agent	Performance Information Source
0	Fixed raw performance values for a discrete set of predefined, feasible, and acceptable alternatives	Measurements, calculations, catalogues, expert estimates, historical data, or outputs from external models
1	Generic analytical performance functions evaluated for variable design inputs	Established mathematical, physical, engineering, economic, or social-science equations and relationships
2	Enterprise-specific adaptations, reduced-order, empirical, statistical, or surrogate performance models	Enterprise data, experiments, calibration, domain expertise, or outputs from Level 3 models
3	Detailed behavioural or simulation-based performance models evaluated for variable design inputs	Specialist engineering, simulation, operational, digital-twin, or enterprise performance software

Methodological Positioning

IMAP → Preferendus-IMAP-IGS

IMAP combines A-FINE preference aggregation with the IGS solver family to identify the preferred best-fit solution within a design space.

Comparable to: MIP, CP, CPLEX, and evolutionary optimisation methods.

Comparative advantage: *For preference-performance-driven design problems, IMAP can outperform conventional mathematical programming and evolutionary optimisation approaches in identifying the best-fit-for-common-purpose solution, because it directly maximises decision-valid aggregated preferences rather than separate objectives or surrogate performance measures*.

TOMAP → Preferendus-TOMAP

TOMAP does not search for an optimum. It directly evaluates alternatives through A-FINE aggregated preferences, revealing explicit trade-offs and producing a unique preference-based ranking for decision support.

Comparable to: TOPSIS, MAUT, AHP, PROMETHEE, and ELECTRE.

Comparative advantage: *For multi-attribute, multi-actor trade-off problems, TOMAP generates a unique, preference-consistent ranking of alternatives by directly aggregating decision-valid preferences within a unified Linear Preference Space. Unlike conventional MCDA approaches, it preserves preference meaning across criteria and actors, providing a mathematically sound basis for ranking, trade-off analysis, and decision support*.

ODESYS-in-FIVES → Preferendus-Agent

Preferendus-Agent operationalizes the broader ODESYS methodology. As an intelligent orchestration layer, it supports the formulation, analysis, and solution of complex multi-attribute, multi-actor design and decision problems in a mathematically sound and decision-valid manner. It encompasses ODESYS methods and software tools trained in AI-skills.

Outperforming: AI-driven open design and decision support systems, decision-intelligence platforms, and engineering copilots.

Comparative advantage: *Preferendus-Agent distinguishes itself as a state-of-the-art, AI-driven open design and decision support system that keeps human judgement in the loop. Rather than producing an automated numerical ordering without clear meaning, it helps users formulate the right problem of interest and establishes the best possible synthesis between what people prefer and what the system can feasibly deliver. By orchestrating decision-valid preference aggregation, performance modelling, design optimisation, and decision analysis, Preferendus-Agent produces preference-meaningful, transparent, and mathematically sound outcomes for design and decision support*.